

Example A: Sales & Inventory System with Forecasting

1. What are the existing processes and challenges in the company's sales and inventory management system in terms of:

- Recording sales transactions
- Tracking and updating inventory levels
- Generating sales and inventory reports
- Identifying personnel responsible for sales and inventory-related tasks

2. What key features should be developed in a web-based sales and inventory management system with forecasting capabilities in terms of:

- Automating sales and inventory recording
- Enhancing real-time inventory monitoring
- Improving sales and inventory report generation
- Integrating sales forecasting for demand prediction
- Implementing user roles and access control

3. Which software development methodology is most suitable for the development of a web-based sales and inventory management system with forecasting capabilities?

4. Which machine learning model and algorithm are most suitable for forecasting sales and inventory demand based on the company's historical and seasonal data?

5. How acceptable is the developed web-based sales and inventory management system with forecasting capabilities based on the ISO/IEC 25010:2015 software quality standards in terms of:

- Functionality
- Performance efficiency
- Compatibility
- Usability
- Reliability
- Security
- Maintainability
- Portability

Example B: Attendance Monitoring System with RFID & Face Verification

1. What are the existing processes and challenges in the institution's attendance management system in terms of:

- Recording attendance of students and/or employees
- Monitoring attendance records in real time
- Generating attendance and discrepancy reports
- Identifying personnel responsible for attendance-related tasks

2. What key features should be developed in a web-based attendance monitoring system using RFID and face verification technology in terms of:

- Automating attendance recording through RFID scanning
- Verifying identity using face recognition to prevent proxy attendance
- Enhancing attendance monitoring capabilities
- Improving attendance and discrepancy report generation
- Implementing user roles and access control

3. Which software development methodology is most suitable for the development of a web-based attendance monitoring system using RFID and face verification technology?

4. Which machine learning model and algorithm are most suitable for accurately identifying and verifying individuals based on facial recognition data captured during attendance logging?

5. How acceptable is the developed web-based attendance monitoring system using RFID and face verification technology based on the ISO/IEC 25010:2015 software quality standards in terms of:

- Functionality
- Performance efficiency
- Compatibility
- Usability
- Reliability
- Security
- Maintainability
- Portability

Example C: Alumni Employability Model with Classification and Sentiment Analysis

1. What are the existing processes and challenges in the institution's alumni tracking and employability assessment system in terms of:

- Collecting and managing alumni data
- Monitoring employment status and career progression
- Gathering feedback from alumni and employers
- Generating employability and institutional performance reports

2. What key features should be developed in a web-based alumni employability model with tracer and feedback systems utilizing classification and sentiment analysis algorithms in terms of:

- Automating alumni data collection and updating
- Tracking employment outcomes and career trajectories
- Classifying alumni into employability categories
- Analyzing qualitative feedback using sentiment analysis
- Providing real-time reporting for institutional assessment

3. Which software development methodology is most suitable for the development of a web-based alumni employability model with tracer and feedback systems?

4. Which machine learning models and algorithms are most suitable for:

- Classifying alumni employability levels based on academic and employment data
- Performing sentiment analysis on alumni and employer feedback to inform institutional improvements

5. How acceptable is the developed web-based alumni employability model with tracer and feedback systems utilizing classification and sentiment analysis algorithms based on the ISO/IEC 25010:2015 software quality standards in terms of:

- Functionality
- Performance efficiency
- Compatibility
- Usability
- Reliability
- Security
- Maintainability
- Portability